19.3.1 TAxCTL Register

Timer_Ax Control Register

Figure 19-15. TAxCTL Register

15	14	13	12	11	10	9	8
Reserved						TASSEL	
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0
7	6	5	4	3	2	1	0
ID		MC		Reserved	TACLR	TAIE	TAIFG
rw-0	rw-0	rw-0	rw-0	rw-0	w-0	rw-0	rw-0

Table 19-4. TAxCTL Register Description

Bit	Field	Type	Reset	Description
15-10	Reserved	RW	0h	Reserved
9-8	TASSEL	RW	0h	Timer_A clock source select 00b = TAxCLK 01b = ACLK 10b = SMCLK 11b = INCLK
7-6	ID	RW	0h	Input divider. These bits along with the TAIDEX bits select the divider for the input clock. 00b = /1 01b = /2 10b = /4 11b = /8
5-4	MC	RW	0h	Mode control. Setting MCx = 00h when Timer_A is not in use conserves power. 00b = Stop mode: Timer is halted 01b = Up mode: Timer counts up to TAxCCR0 10b = Continuous mode: Timer counts up to 0FFFFh 11b = Up/down mode: Timer counts up to TAxCCR0 then down to 0000h
3	Reserved	RW	0h	Reserved
2	TACLR	RW	0h	Timer_A clear. Setting this bit resets TAxR, the timer clock divider logic, and the count direction. The TACLR bit is automatically reset and is always read as zero.
1	TAIE	RW	0h	Timer_A interrupt enable. This bit enables the TAIFG interrupt request. 0b = Interrupt disabled 1b = Interrupt enabled
0	TAIFG	RW	0h	Timer_A interrupt flag 0b = No interrupt pending 1b = Interrupt pending

19.3.2 TAxR Register

Timer_Ax Counter Register

Figure 19-16. TAxR Register

15	14	13	12	11	10	9	8
TAxR							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0
7	6	5	4	3	2	1	0
TAxR							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 19-5. TAxR Register Description

Bit	Field	Type	Reset	Description
15-0	TAxR	RW	0h	Timer_A register. The TAxR register is the count of Timer_A.

19.3.3 TAxCCTL0 to TAxCCTL6 Register

Timer_Ax Capture/Compare Control 0 Register to Timer_Ax Capture/Compare Control 6

Figure 19-17. TAxCCTL0 to TAxCCTL6 Register

15	14	13	12	11	10	9	8
CM		CCIS		SCS	SCCI	Reserved	CAP
rw-0	rw-0	rw-0	rw-0	rw-0	r-0	r-0	rw-0
7	6	5	4	3	2	1	0
OUTMOD			CCIE	CCI	OUT	COV	CCIFG
rw-0	rw-0	rw-0	rw-0	r	rw-0	rw-0	rw-0

Table 19-6. TAxCCTL0 to TAxCCTL6 Register Description

Bit	Field	Type	Reset	Description
15-14	CM	RW	0h	Capture mode 00b = No capture 01b = Capture on rising edge 10b = Capture on falling edge 11b = Capture on both rising and falling edges
13-12	CCIS	RW	0h	Capture/compare input select. These bits select the TAxCCR0 input signal. See the device-specific data sheet for specific signal connections. 00b = CCIxA 01b = CCIxB 10b = GND 11b = VCC
11	SCS	RW	0h	Synchronize capture source. This bit is used to synchronize the capture input signal with the timer clock. 0b = Asynchronous capture 1b = Synchronous capture
10	SCCI	RW	0h	Synchronized capture/compare input. The selected CCI input signal is latched with the EQUx signal and can be read via this bit.
9	Reserved	R	0h	Reserved. Reads as 0.
8	CAP	RW	0h	Capture mode 0b = Compare mode 1b = Capture mode
7-5	OUTMOD	RW	0h	Output mode. Modes 2, 3, 6, and 7 are not useful for TAxCCR0 because EQUx = EQU0. 000b = OUT bit value 001b = Set 010b = Toggle/reset 011b = Set/reset 100b = Toggle 101b = Reset 110b = Toggle/set 111b = Reset/set
4	CCIE	RW	0h	Capture/compare interrupt enable. This bit enables the interrupt request of the corresponding CCIFG flag. 0b = Interrupt disabled 1b = Interrupt enabled
3	CCI	R	0h	Capture/compare input. The selected input signal can be read by this bit.
2	OUT	RW	0h	Output. For output mode 0, this bit directly controls the state of the output. 0b = Output low 1b = Output high
1	COV	RW	0h	Capture overflow. This bit indicates a capture overflow occurred. COV must be reset with software. 0b = No capture overflow occurred 1b = Capture overflow occurred
0	CCIFG	RW	0h	Capture/compare interrupt flag 0b = No interrupt pending 1b = Interrupt pending

19.3.4 TAxCCR0 to TAxCCR6 Register

Timer_Ax Capture/Compare 0 Register to Timer_Ax Capture/Compare 6 Register

Figure 19-18. TAxCCR0 to TAxCCR6 Register

15	14	13	12	11	10	9	8
TAxCCRn							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0
7	6	5	4	3	2	1	0
TAxCCRn							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 19-7. TAxCCR0 to TAxCCR6 Register Description

Bit	Field	Type	Reset	Description
15-0	TAxCCR0	RW	0h	Compare mode: TAxCCRn holds the data for the comparison to the timer value in the Timer_A Register, TAxR. Capture mode: The Timer_A Register, TAxR, is copied into the TAxCCRn register when a capture is performed.

19.3.5 TAxIV Register

Timer_Ax Interrupt Vector Register

Figure 19-19. TAxIV Register

15	14	13	12	11	10	9	8
TAIV							
r0	r0	r0	r0	r0	r0	r0	r0
7	6	5	4	3	2	1	0
TAIV							
r0	r0	r0	r0	r-0	r-0	r-0	r0

Table 19-8. TAxIV Register Description

Bit	Field	Type	Reset	Description
15-0	TAIV	R	0h	Timer_A interrupt vector value 00h = No interrupt pending 02h = Interrupt Source: Capture/compare 1; Interrupt Flag: TAxCCR1 CCIFG; Interrupt Priority: Highest 04h = Interrupt Source: Capture/compare 2; Interrupt Flag: TAxCCR2 CCIFG 06h = Interrupt Source: Capture/compare 3; Interrupt Flag: TAxCCR3 CCIFG 08h = Interrupt Source: Capture/compare 4; Interrupt Flag: TAxCCR4 CCIFG 0Ah = Interrupt Source: Capture/compare 5; Interrupt Flag: TAxCCR5 CCIFG 0Ch = Interrupt Source: Capture/compare 6; Interrupt Flag: TAxCCR6 CCIFG 0Eh = Interrupt Source: Timer overflow; Interrupt Flag: TAxCTL TAIFG; Interrupt Priority: Lowest

19.3.6 TAxEX0 Register

Timer_Ax Expansion 0 Register

Figure 19-20. TAxEX0 Register

15	14	13	12	11	10	9	8
Reserved							
r0	r0	r0	r0	r0	r0	r0	r0
7	6	5	4	3	2	1	0
Reserved					TAIDEX[1]		
r0	r0	r0	r0	r0	rw-0	rw-0	rw-0

(1) After programming TAIDEX bits and configuration of the timer, set TACLR bit to ensure proper reset of the timer divider logic.

Table 19-9. TAxEX0 Register Description

Bit	Field	Type	Reset	Description
15-3	Reserved	R	0h	Reserved. Reads as 0.
2-0	TAIDEX	RW	0h	Input divider expansion. These bits along with the ID bits select the divider for the input clock. 000b = Divide by 1 001b = Divide by 2 010b = Divide by 3 011b = Divide by 4 100b = Divide by 5 101b = Divide by 6 110b = Divide by 7 111b = Divide by 8